

National University of Singapore
School of Computing

BT2102 Data Management and Visualisation

Tutorial 5:
Data Visualisation

1. Critique the following visualisations.

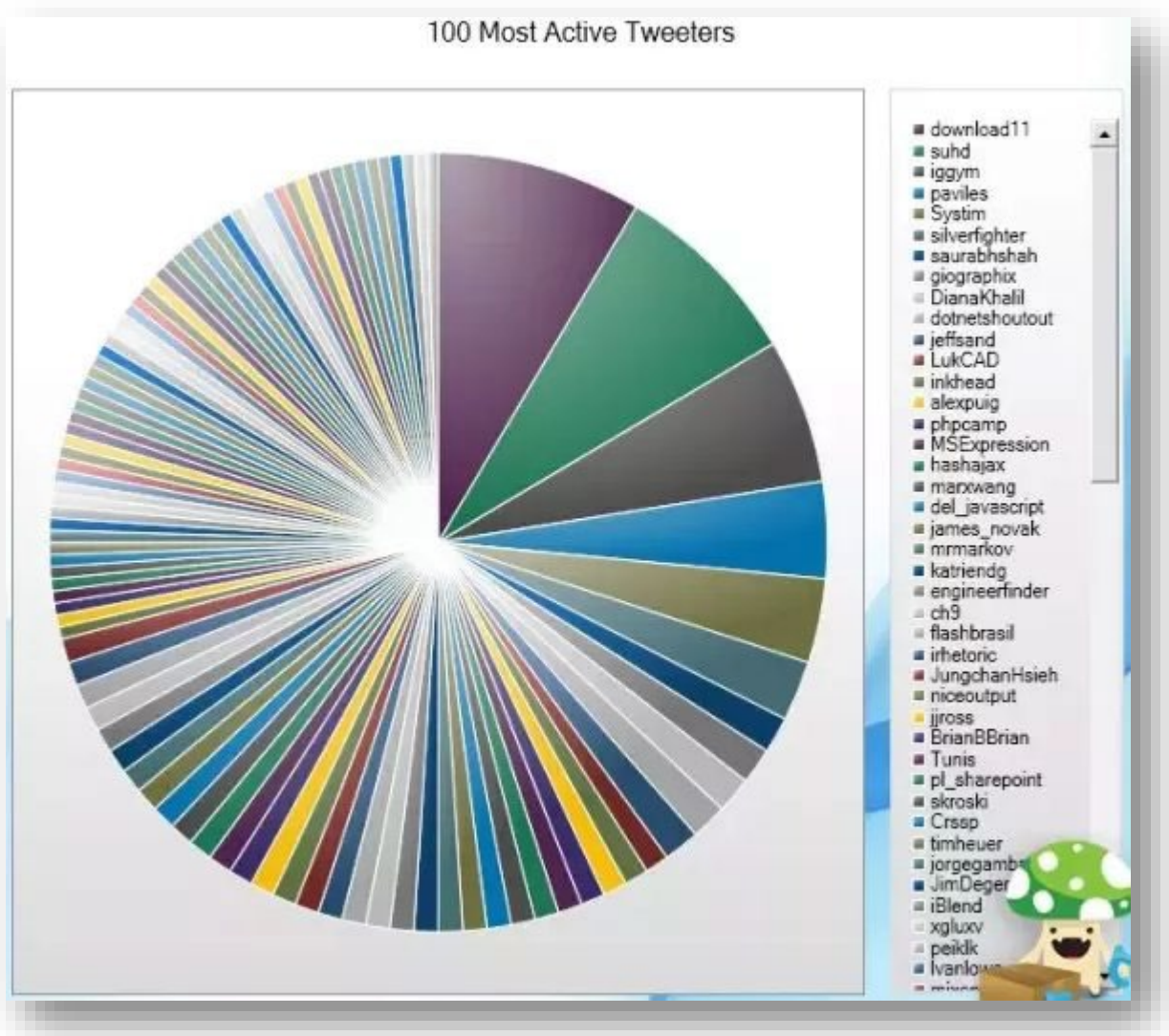
a)



b)

Category	Year of Ord..	Order Date											
		January	February	March	April	May	June	July	August	September	October	November	December
Furniture	2014	\$6,243	\$1,840	\$14,574	\$7,945	\$6,913	\$13,206	\$10,821	\$7,320	\$23,816	\$12,304	\$21,565	\$30,646
	2015	\$11,740	\$3,134	\$12,500	\$10,476	\$9,375	\$7,714	\$13,674	\$9,639	\$26,273	\$12,027	\$30,881	\$23,086
	2016	\$7,623	\$3,926	\$12,801	\$13,212	\$15,120	\$13,071	\$13,069	\$12,483	\$27,263	\$11,873	\$31,784	\$36,679
	2017	\$5,964	\$6,866	\$10,893	\$9,066	\$16,958	\$19,009	\$11,813	\$15,442	\$29,028	\$21,884	\$37,057	\$31,407
Office Supplies	2014	\$4,851	\$1,072	\$8,606	\$11,155	\$7,136	\$12,953	\$15,121	\$11,379	\$27,423	\$7,211	\$26,862	\$18,006
	2015	\$1,809	\$5,368	\$15,883	\$12,559	\$9,114	\$10,648	\$4,720	\$11,735	\$19,306	\$8,673	\$21,218	\$16,202
	2016	\$5,300	\$6,794	\$17,347	\$10,647	\$13,035	\$10,902	\$12,924	\$8,960	\$23,264	\$16,282	\$20,487	\$37,998
	2017	\$21,274	\$7,408	\$14,550	\$15,072	\$13,737	\$16,912	\$10,241	\$30,060	\$31,896	\$23,037	\$31,472	\$30,437
Technology	2014	\$3,143	\$1,609	\$32,511	\$9,195	\$9,600	\$8,436	\$8,004	\$9,210	\$30,538	\$11,938	\$30,201	\$20,893
	2015	\$4,625	\$3,449	\$10,344	\$11,161	\$11,643	\$6,435	\$10,371	\$15,525	\$19,017	\$10,705	\$23,874	\$35,632
	2016	\$5,620	\$12,259	\$21,568	\$14,891	\$28,833	\$16,372	\$13,269	\$9,672	\$22,883	\$31,533	\$27,141	\$22,323
	2017	\$16,733	\$6,027	\$33,429	\$12,383	\$13,567	\$17,061	\$23,210	\$17,619	\$26,943	\$32,856	\$49,919	\$21,985

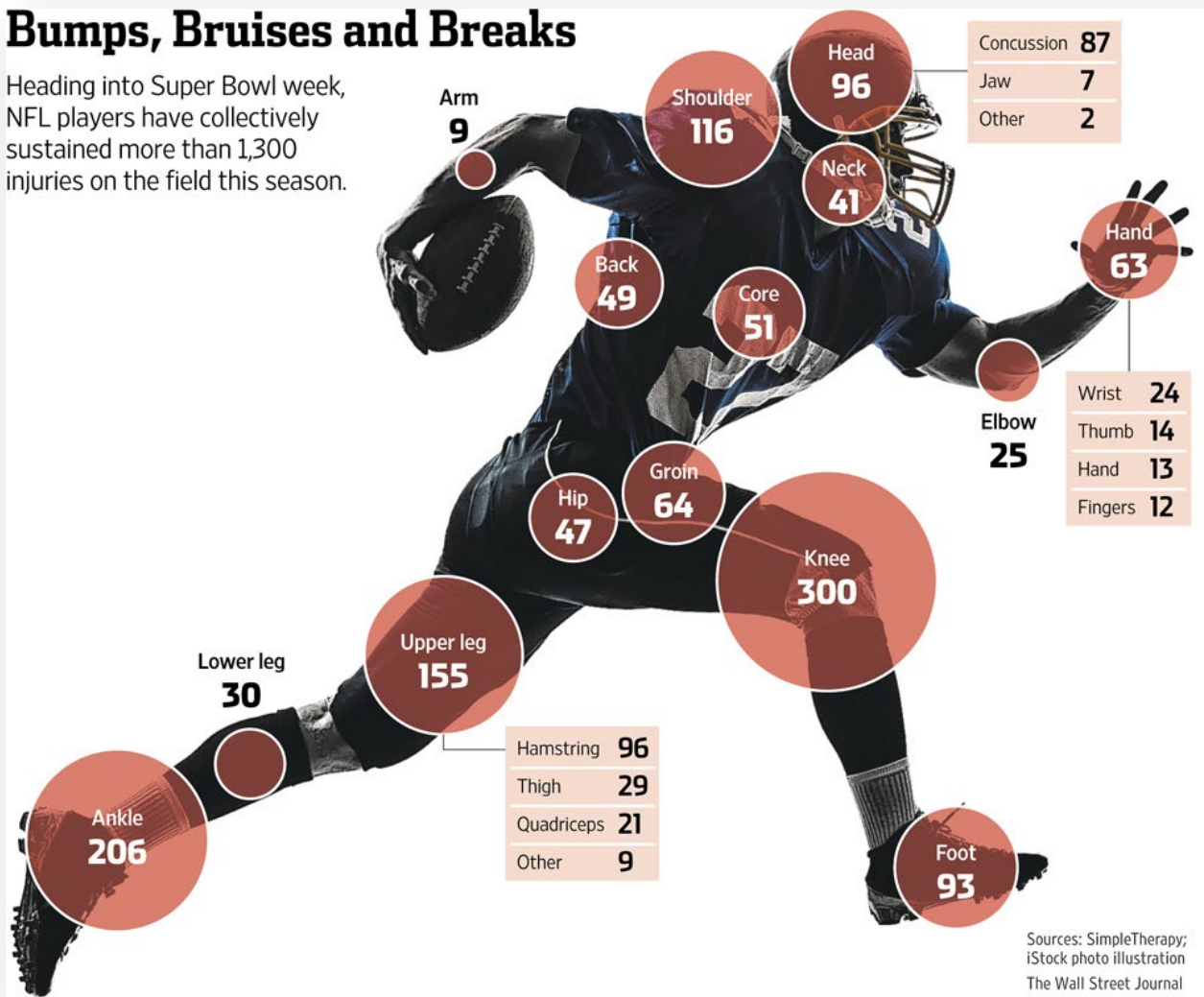
c)



d)

Bumps, Bruises and Breaks

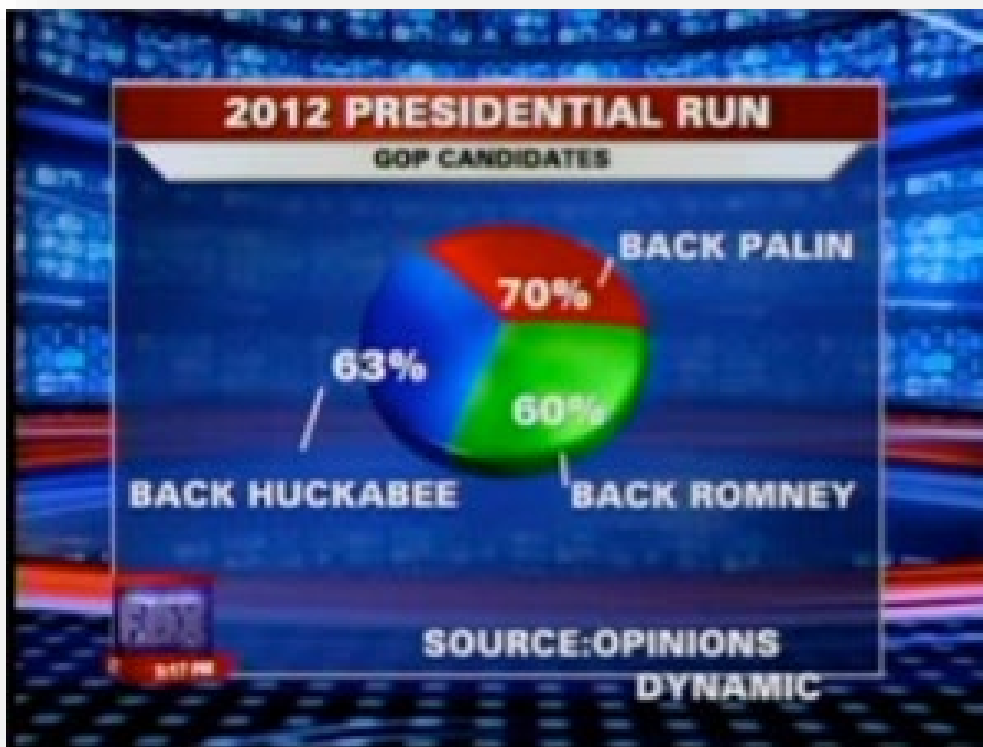
Heading into Super Bowl week, NFL players have collectively sustained more than 1,300 injuries on the field this season.



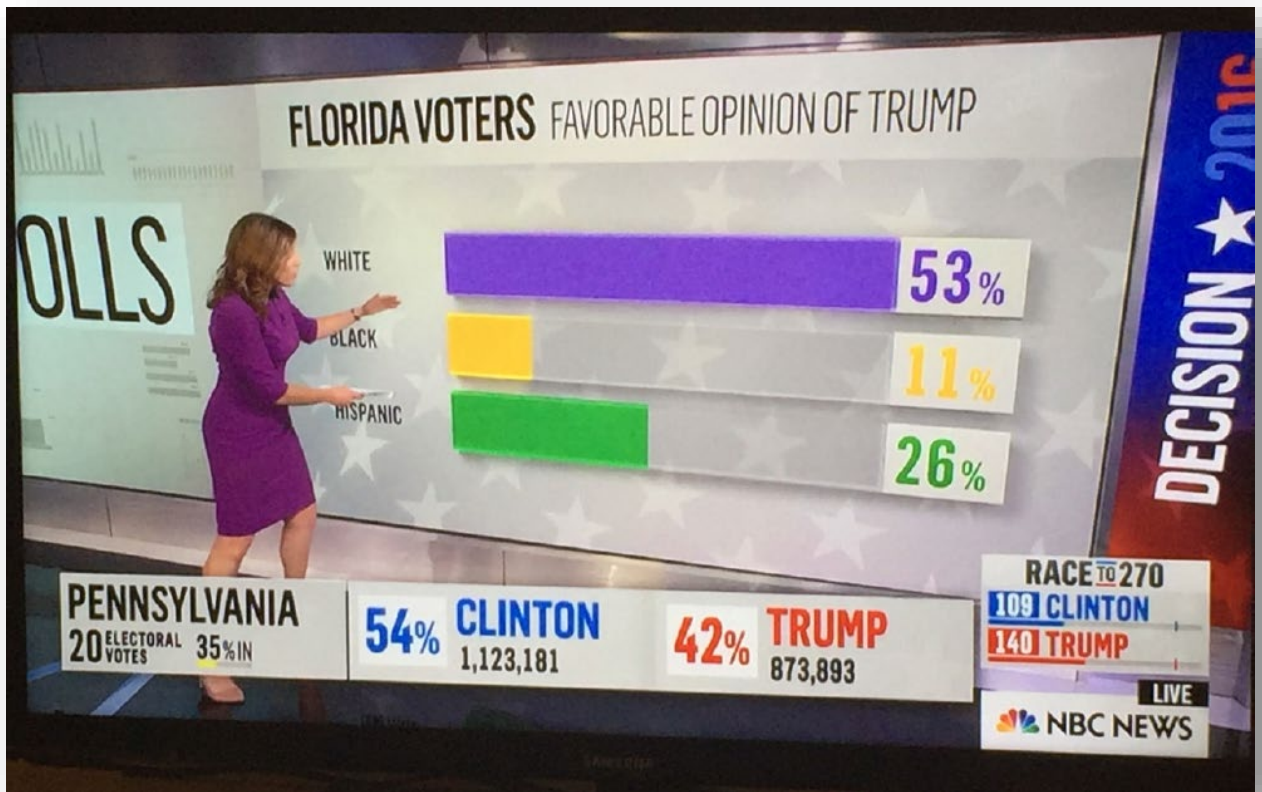
e)



f)



g)



2.

To work in the USA, a foreigner must complete several applications. An important one is to apply to the Immigration service for the appropriate visa (green card, H1B, etc.). Before this can be done, the company that intends to hire him must have the candidate certified by the Department of Labor (DOL).

The file “Jobs.xlsx” is a compiled set of 160,000+ applications for certification to the US DOL from 2008 to 2015. The applications include the proposed salary for the foreign worker, and the prevailing salary for a local candidate (this is called the prevailing wage). The DOL checks to ensure equity between foreigners and locals. At the end of this application, the company’s application will be ‘certified’ or ‘denied’ by the DOL. If it is certified, the candidate can proceed to apply for his visa.

Here are the fields in the CSV file:

Case Number:	a unique identifier for each application
Case Status:	the last significant event in the case. Values include “Certified”, “Certified Expired”, “Denied” and “Withdrawn”
Case Received Date:	the date the application was received
Decision Date:	date of the last significant event
Employer Name:	the name of the employing company applying for certification
Prevailing Wage Submitted:	the prevailing wage of the job being offered to a foreigner
Prevailing Wage Submitted Unit:	the unit of pay. Valid values are hourly, weekly, biweekly, monthly, yearly
Paid Wage Submitted:	the lower range of the wage offered to the foreign applicant
Paid Wage Submitted Unit:	the unit of pay. Valid values are hourly, weekly, biweekly, monthly, yearly
Job Title:	title of the job being offered
Work City:	the foreign worker's intended area of employment (city)
Education Level Required:	minimum level of education required
College Major Required:	field of study required based on the education requirement
Experience Required:	whether experience in the job offered is a requirement (Y/N).
Experience Required Number of Months:	number of months experience that are required
Country of Citizenship:	country of citizenship of the foreign worker
Prevailing Wage Soc Code:	occupational code (SOC) associated with the job being requested
Prevailing Wage Soc Title:	name associated with the SOC Code
Work State:	the foreign worker's intended area of employment (state)
Work Postal Code:	the foreign worker's intended area of employment (zip code)
Full Time Position:	whether the position is full-time (Y) or part-time (N).
Visa Class:	type of application submitted for processing. Options include “H-1B,” “E-3 Australian,” “H-1B1 Chile,” “H-1B1 Singapore,” “greencard.”
Job Title Subgroup:	a more generic categorisation label created based on the Job Title field. Options include assistant professor, attorney, business analyst, data analyst , data scientist , management consultant, software engineer, and teacher.

Your Tasks:

There are two job types that are of interest to us in this analysis: **data analyst** and **data scientist**.

The given dataset has job titles that are associated with these two job types (**such as data scientist, associate data scientist, senior data scientist, data analyst, business data analyst, clinical data analyst, statistical data analyst, etc.**) Using the data provided, explore, and identify the job titles, locations, companies and salaries of these **data-related jobs**, present insights you have gathered from your exploration to your peers visually and explain your representation/presentation design decisions.

You could consider the level-1 question to be “**What is the prospect of data analyst and data scientist in the USA given the dataset?**”

You may consider some of these deeper level questions before deciding how to present the insights to your audience:

1. How do these two data-related job types compare with other job types (such as attorney, software engineer, teacher, etc.)?
2. Are there any salary differences between these two data-related job types based on the type of visa application?
3. How do the two data-related job types compare with the other job types with respect to the visa application?
4. How do the annual wages paid (for all job types) compare across the different states?
5. How do the offered salaries of those visa holders (i.e., non-green card) compare with the prevailing wages of the permanent residents (i.e., those holding green cards) in the two data-related jobs?

You will be asked to show your data visualisations and explain your data insights including recommendations where applicable during the tutorial session.